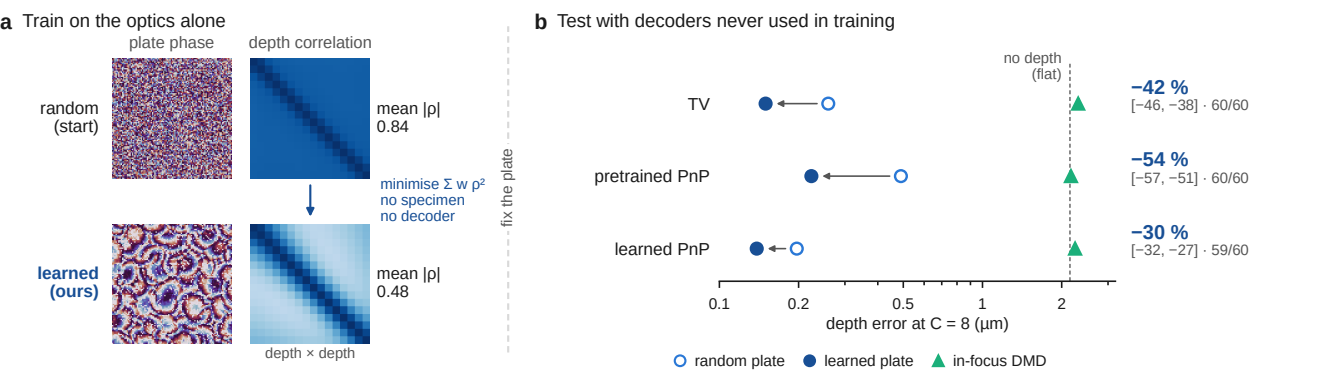

a Train on the optics alone
plate phase
depth correlation
random (start)
mean |ρ| 0.84
minimise Σ w ρ²
no specimen
no decoder
learned (ours)
mean |ρ| 0.48
depth × depth
fix the plate
b Test with decoders never used in training
no depth (flat)
TV
−42 %
[−46, −38] · 60/60
pretrained PnP
−54 %
[−57, −51] · 60/60
learned PnP
−30 %
[−32, −27] · 59/60
0.1
0.2
0.5
1
2
depth error at C = 8 (μm)
random plate
learned plate
in-focus DMD